Geometry — Name Answer Key
Unit 12 — Date______________________
Lesson 8 Homework — Period______________________

For questions 1 and 2, consider the equation of the circle represented by
$(\mathrm{x}+3)^2+(\mathrm{y}+4)^2=9^2$.

1. Rewrite the equation in general form.

$$x^2+6x+9+y^2+8y+16=81$$
$$x^2+y^2+6x+8y-56=0$$

2. Give the key features of the circle.

<table>
<tr><th>Center</th><th>Diameter</th><th>Radius</th></tr>
<tr><td>$(-3,-4)$</td><td>18</td><td rowspan="3">9</td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td>$-12 \le x \le 6$</td><td>$-13 \le y \le 5$</td></tr>
</table>

For questions 3 and 4, consider the equation of the circle represented by
$(\mathrm{x}-5)^2+(\mathrm{y}+2)^2=5^2$.

3. Rewrite the equation in general form.

$$x^2-10x+25+y^2+4y+4=25$$
$$x^2+y^2-10x+4y+4=0$$

4. Give the key features of the circle.

<table>
<tr><th>Center</th><th>Diameter</th><th>Radius</th></tr>
<tr><td>$(5,-2)$</td><td>10</td><td rowspan="3">5</td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td>$0 \le x \le 10$</td><td>$-7 \le y \le 3$</td></tr>
</table>

For questions 5 and 6, consider the equation of the circle represented by
$x^2 + y^2 - 2x - 6y = -6$.

5. Rewrite the equation in center-radius form.

$$x^2 - 2x + y^2 - 6y = -6$$
$$x^2 - 2x + 1 + y^2 - 6y + 9 = -6 + 1 + 9$$
$$(x-1)^2 + (y-3)^2 = 4$$

6. Give the key features of the circle.

Center	Diameter	Radius
(1,3)	4	2
Domain	**Range**	
$-1 \leq x \leq 3$	$1 \leq y \leq 5$	

For questions 7 and 8, consider the equation of the circle represented by
$x^2 + y^2 - 20y = 21$.

7. Rewrite the equation in center-radius form.

$$x^2 + y^2 - 20y = 21$$
$$x^2 + y^2 - 20y + 100 = 21 + 100$$
$$x^2 + (y-10)^2 = 121$$

8. Give the key features of the circle.

Center	Diameter	Radius
(0,10)	22	11
Domain	**Range**	
$-11 \leq x \leq 11$	$-1 \leq y \leq 21$	

For questions 9 and 10, consider the equation of the circle represented by
$x^2 + y^2 - 10x = -24$.

9. Rewrite the equation in center-radius form.

$$x^2 + y^2 - 10x = -24$$
$$x^2 - 10x + y^2 = -24$$
$$x^2 - 10x + 25 + y^2 = -24 + 25$$
$$(x-5)^2 + y^2 = 1$$

10. Give the key features of the circle.

Center	Diameter	Radius
(5,0	2	1
Domain	**Range**	
$4 \leq x \leq 6$	$-1 \leq y \leq 1$	